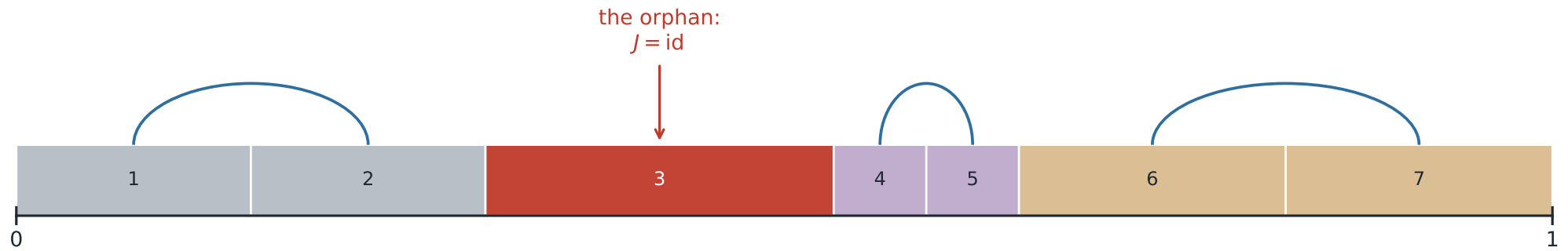
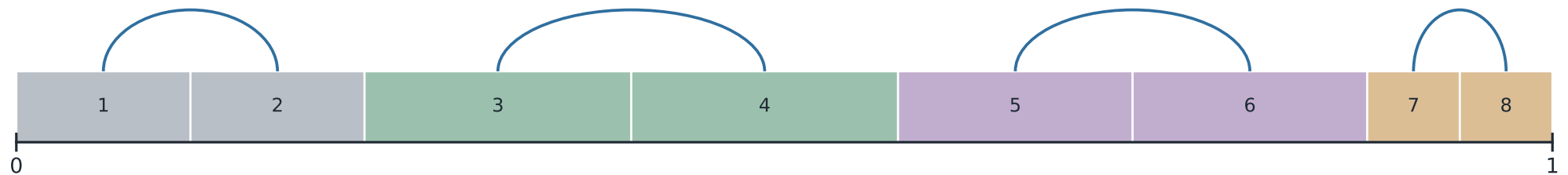


(a) DATA — $2\alpha/\pi = 5/9$, P odd: $n = 7$, three mirror pairs (arcs) of equal width and one orphan.



(b) DATA — $2\alpha/\pi = 2/9$, P even: $n = 8$, four mirror pairs and no orphan.



Branch endpoints are exact (lib/partition.py, in $\mathbb{Z}[\zeta_4\mathbb{Q}]$); the arcs are the first-return map J , computed by tracing each branch to its perpendicular return, so the equality of paired widths is an observation here and not a definition.